

JIM CAMPBELL

Field Service Engineer

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Portfolio: sites.google.com/view/idevdroidapps

SUMMARY

Robotics Field Service Engineer with a B.S. in Computer Engineering and 20+ years of experience in the deployment, commissioning, and on-site maintenance of autonomous systems. Expert in hardware diagnostics, component-level repair, and electrical troubleshooting (CAN bus, sensors, power systems) for mission-critical robotics and automotive platforms. Proven track record of leading international field operations, performing site surveys, and managing complex system bring-ups to ensure maximum uptime in demanding environments.

SKILLS

Field Operations & Deployment

- **Site Commissioning:** Lead On-site Installation, SAT/FAT (Site/Factory Acceptance Testing), and System Bring-Up.
- **Field Maintenance:** Expert in Preventive/Corrective Maintenance, Failure Recovery, and Hardware Lifecycle Support.
- **Logistics & Safety:** International Field Operations, Site Surveys, and Safety System Validation (E-Stop, Redundancy).
- **Documentation:** Technical Field Reporting, Root Cause Analysis (RCA), and Operator Training.

Hardware & Electrical Diagnostics

- **Diagnostic Tools:** Proficient with Multimeters, Oscilloscopes, Logic Analyzers, and Spectrum Analyzers.
- **Prototyping & Repair:** Component-Level Troubleshooting, PCB Trace Repair, and JTAG/SWD Debugging.
- **Electrical Systems:** Wiring Harness Diagnostics, Signal Integrity, Thermal Analysis, and Load Testing.

Robotics & Electromechanical Systems

- **Motion Control:** DC Motor Systems, PWM Control, and PID Loop Tuning/Optimization.

- **Sensor Calibration:** Integration and calibration of LIDAR, GPS, RF, and Environmental Sensors.
- **Navigation Hardware:** SLAM-Based Navigation hardware integration and field-tuning.

Industrial Communication & Networking

- **Vehicle & Industrial Bus:** CAN Bus (J1939, CANopen), RS-232/485, I2C, and SPI.
 - **Field Networking:** TCP/IP Networking, RF Communication Systems, and GPS Telemetry.
 - **Linux Diagnostics:** Linux CLI for System Monitoring, Shell Scripting, and Serial Terminal Tools (PuTTY).
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Professional Experience

Independent Freelance

Field Systems Integration & Deployment Engineer | Hollywood, FL | September 2025 – Present

- **On-Site System Deployment:** Managed the physical installation and hardware commissioning of Ubuntu-based edge compute nodes, including rack mounting, cable management, and network infrastructure integration.
- **Hardware Bring-Up & Validation:** Performed end-to-end hardware initialization, including BIOS/UEFI optimization and peripheral stress testing to ensure stability in high-duty-cycle environments.
- **Field Diagnostics & Recovery:** Utilized Linux CLI tools to perform real-time hardware health monitoring, identifying and mitigating thermal throttling and power distribution issues to maintain ~99% uptime.
- **Preventive Maintenance:** Executed scheduled hardware audits, log analysis, and physical inspections to identify wear-and-tear in deployed infrastructure before failure.
- **Infrastructure Integration:** Configured local storage arrays and hardware-level data persistence layers to ensure system reliability in offline or intermittent-connectivity environments.

Rivian & Volkswagen Group Technologies Joint Venture

Field Systems Integration Engineer (Automotive Platforms) | Remote / Field Testing | March 2025 – September 2025

- **Vehicle System Integration:** Supported integration and validation of critical vehicle systems on Android Automotive OS platforms, working directly with pre-production hardware in field and test environments.

- **Hardware-Level Diagnostics:** Diagnosed and resolved system failures on vehicle hardware platforms, including communication faults, performance bottlenecks, and subsystem integration issues, improving system stability by 10–20%.
- **Field Testing & Validation:** Executed on-vehicle testing and validation procedures to ensure reliable operation of navigation, sensor, and HMI systems under real-world conditions.
- **Production Deployment Support:** Collaborated with cross-functional teams to support early-stage vehicle production and field deployments, ensuring successful system bring-up and configuration.
- **Communication Protocol Analysis:** Validated and troubleshoot CAN bus and other vehicle communication interfaces to ensure consistent data flow between embedded controllers and system components.
- **System Reliability Engineering:** Identified root causes of system instability and implemented corrective actions to improve uptime and reduce critical system failures in field testing environments.

Canoo Inc.

Lead Systems Integration Engineer (Vehicle Platforms) | Remote / Field Operations | March 2024 – January 2025

- **Vehicle Systems Standardization:** Led the development of unified communication frameworks to standardize sensor data and control signals across vehicle platforms, reducing hardware integration conflicts by ~30–40%.
- **Deployment Efficiency Improvement:** Improved system modularity and configuration consistency, enabling faster and more reliable hardware bring-up and feature deployment in vehicle platforms (~20–30% improvement).
- **Hardware-Software Integration Oversight:** Directed cross-functional efforts to resolve system-level integration issues between embedded controllers, sensors, and vehicle subsystems.
- **Field Diagnostics Leadership:** Guided troubleshooting strategies for complex vehicle system failures, supporting rapid identification and resolution of hardware communication and control issues.
- **Technical Mentorship:** Trained and mentored engineers on system diagnostics, integration workflows, and best practices for maintaining stability in field-deployed vehicle systems.

Senior Systems Integration Engineer (Vehicle Platforms) | Remote / Field Operations | April 2023 – March 2024

- **Vehicle Diagnostics Interface Development:** Developed and deployed a real-time diagnostic interface for monitoring vehicle subsystems, enabling rapid identification of hardware faults and system performance issues.

- **Hardware-Level Troubleshooting:** Diagnosed and resolved critical communication failures across vehicle platforms using CAN bus analysis and low-level system diagnostics, improving system uptime and sensor reliability.
- **Subsystem Integration & Validation:** Supported integration and validation of vehicle sensors and control modules, ensuring reliable communication between embedded systems in pre-production environments.
- **Telemetry-Based System Monitoring:** Implemented real-time telemetry solutions to track hardware performance and detect anomalies, enabling proactive maintenance and faster failure resolution.
- **Field Support for Vehicle Systems:** Assisted in troubleshooting and stabilizing vehicle platforms during field operations and testing, ensuring consistent system behavior under real-world conditions.

Milwaukee Electric Tool Corporation

Principal Systems Integration Engineer (Embedded Hardware) | Remote | January 2022 – April 2022

- **Hardware Feature Integration:** Led cross-functional efforts to integrate embedded systems with physical power tool platforms, ensuring reliable interaction between firmware, sensors, and electromechanical components.
- **System Validation & Test Support:** Supported validation of hardware features through structured testing and diagnostics, improving system reliability prior to production release.
- **Integration Issue Resolution:** Identified and resolved system-level integration issues between hardware and embedded controllers, reducing rework and improving deployment readiness.
- **Product Reliability Improvement:** Drove initiatives to improve stability and consistency of hardware behavior under real-world operating conditions.
- **Technical Leadership & Mentorship:** Guided engineers on hardware-focused debugging techniques and integration best practices, improving team effectiveness in diagnosing system failures.

Nautilus-Bowflex Inc.

Systems Integration Engineer (Embedded & Sensor Systems) | Remote | June 2021 – January 2022

- **Sensor System Integration:** Integrated camera-based sensing systems with embedded hardware platforms to enable real-time motion tracking and performance feedback.
- **Hardware Validation & Calibration:** Performed calibration and validation of sensor systems against real-world user movements to ensure accuracy and consistent system response.

- **Hardware-in-the-Loop Testing:** Conducted HIL testing to validate system behavior under live operating conditions, ensuring low-latency and reliable interaction between sensors and control systems.
- **Real-Time System Diagnostics:** Monitored and analyzed system performance to identify inconsistencies in sensor data and hardware response, improving overall system reliability.
- **Embedded System Optimization:** Improved system responsiveness and accuracy by tuning interactions between sensor inputs and embedded processing systems.

SharkNinja Inc.

Systems Integration Engineer (Robotic Platforms) | Remote | December 2020 – February 2021

- **Robotic System Integration:** Supported integration of embedded systems within consumer robotic platforms, ensuring reliable coordination between sensors, motor controllers, and onboard processing units.
- **Subsystem Performance Optimization:** Improved system stability and responsiveness by resolving integration issues between hardware components and embedded control systems.
- **Hardware-Software Diagnostics:** Identified and troubleshoot system-level failures affecting real-world device performance, improving operational reliability of deployed products.
- **System Reliability Enhancement:** Reduced hardware-software integration failures by refining system interactions and improving fault handling across robotic subsystems.
- **Technical Leadership:** Coordinated with engineering teams to improve system maintainability and ensure consistent behavior across hardware platforms.

Shield AI

Field Systems Integration Engineer (Autonomous Platforms) | Remote / Field Testing | September 2020 – December 2020

- **Autonomous System Integration:** Supported integration of SLAM-based navigation and sensor systems on autonomous hardware platforms operating in GPS-denied environments.
- **Field Testing & Deployment:** Participated in field testing and system validation, ensuring reliable operation of autonomous platforms under real-world mission conditions.
- **Hardware Communication Diagnostics:** Diagnosed and resolved communication issues between onboard systems and remote operator interfaces, improving reliability of real-time control and telemetry.
- **Operator System Support:** Enabled stable connectivity between autonomous systems and field operators, ensuring consistent situational awareness and control during live operations.

- **Defense-Grade System Reliability:** Contributed to system stability and compliance for ruggedized deployments, ensuring consistent performance in high-demand operational environments.
- **Edge System Synchronization:** Supported real-time data synchronization between mobile hardware units and command systems to maintain accurate and timely system state awareness.

ReachMobi Inc.

Senior Systems Engineer | Remote | December 2018 – July 2020

- **System Reliability Engineering:** Led efforts to improve stability and uptime of high-traffic distributed systems operating in production environments.
- **Infrastructure Scaling:** Expanded engineering capacity and system throughput, supporting increased operational demand and deployment scale.
- **Production Issue Resolution:** Diagnosed and resolved critical system issues impacting live environments, ensuring consistent service availability.
- **Cross-Functional Coordination:** Collaborated with engineering and operations teams to maintain system performance and reliability across distributed platforms.
- **Project Delivery & Lifecycle Management:** Managed full lifecycle delivery of production systems, ensuring consistent and reliable deployment outcomes.

Systems Engineer | Remote | September 2017 – December 2018

- **System Deployment at Scale:** Supported deployment of large-scale distributed systems across global environments, ensuring consistent configuration and reliable operation.
- **System Migration & Stability:** Led migration of core system architecture, maintaining system integrity and minimizing disruption during transition.
- **Cross-System Integration:** Standardized communication between distributed system components to ensure reliable data exchange and low-latency performance.
- **Production Support & Troubleshooting:** Investigated and resolved system issues impacting live environments, improving overall system stability and uptime.

Televisa Internacional LLC

Embedded Systems Engineer | Miami, FL | February 2016 – August 2017

- **Embedded Device Integration:** Supported deployment of software systems across a wide range of consumer hardware platforms, ensuring compatibility and consistent performance.
- **Hardware Compatibility & Validation:** Tested and validated system behavior across diverse device configurations, identifying and resolving hardware-specific issues.

- **System Performance Optimization:** Improved system responsiveness and stability on resource-constrained embedded devices, ensuring reliable operation under production conditions.
- **Production Support & Troubleshooting:** Diagnosed and resolved device-level issues impacting system performance in live environments, maintaining high service availability.

Independent Freelance

Field Service Systems Integrator (Commercial Kiosk Platforms) | Hollywood, FL | July 2015 – January 2016

- **On-Site Hardware Installation:** Performed end-to-end installation and commissioning of commercial kiosk systems in retail environments, including physical setup, wiring, and system configuration.
- **System Integration:** Integrated kiosk hardware with backend infrastructure to enable real-time transaction processing and inventory management.
- **Site Surveys & Deployment Preparation:** Assessed installation sites to determine power, network, and mounting requirements, ensuring successful system deployment.
- **Connectivity & Power Configuration:** Configured network connectivity and power systems to ensure stable and continuous operation of field-deployed hardware.
- **Post-Deployment Support:** Provided on-site troubleshooting and maintenance to resolve hardware and connectivity issues, ensuring reliable system performance.

IC Realtime LLC

Systems Integration Engineer (Security & Surveillance Systems) | Pompano Beach, FL | March 2013 – June 2015

- **Multi-System Hardware Integration:** Integrated surveillance cameras, biometric access control systems, and mobile interfaces to enable centralized monitoring and control of physical security infrastructure.
- **Low-Latency System Control:** Engineered and optimized communication between client devices and remote hardware systems, achieving 2–3 second response times for real-time control and monitoring.
- **Multi-Vendor System Integration:** Integrated hardware and software components across multiple manufacturers, ensuring compatibility and reliable operation within complex deployment environments.
- **System Commissioning & Validation:** Performed technical validation of deployed systems, verifying correct operation of hardware, network communication, and control interfaces prior to handoff.
- **Deployment Support & Troubleshooting:** Diagnosed and resolved integration and performance issues affecting live systems, improving reliability in production environments.

Field Support Engineer | Pompano Beach, FL | March 2013 – June 2015

- **On-Site Installation & Commissioning:** Installed and configured surveillance and biometric access control systems in commercial environments, including wiring, mounting, and system bring-up.
- **Field Diagnostics & Repair:** Served as the primary on-site technical resource for diagnosing and resolving hardware, wiring, and network issues in complex system deployments.
- **Systems Integration:** Integrated security hardware with automation platforms (Crestron, Control4, Savant), ensuring seamless communication between subsystems.
- **Customer-Facing Technical Support:** Worked directly with clients and site operators to resolve issues, provide system training, and ensure successful deployment outcomes.
- **Maintenance & Service Operations:** Performed ongoing service, upgrades, and repairs on deployed systems to maintain reliability and long-term performance.

FutureGenRobotics LLC

Embedded Systems Engineering Intern (Robotics Systems) | Boca Raton, FL | January 2012 – December 2012

- **Autonomous System Integration:** Supported development and integration of an autonomous robotic platform, coordinating communication between sensors, controllers, and navigation systems.
- **Motor Control & Drive Systems:** Implemented and tuned BLDC motor control systems, improving torque efficiency and overall system performance by ~15–20%.
- **Sensor Integration & Calibration:** Integrated and calibrated GPS-based positioning systems, improving navigation accuracy and reliability in real-world operation.
- **Safety System Development:** Developed and validated a hardware-based emergency stop system using capacitive sensing, achieving sub-50ms response time for operator safety.
- **Hardware Prototyping & Testing:** Assisted in rapid prototyping and iterative testing of embedded hardware systems, reducing integration issues and improving system stability.

Uplink Corporation

Field Service Engineer (RF & GPS Systems) | Southeast U.S., LATAM, Caribbean Territories | May 2006 – February 2008

- **Regional Field Deployment:** Executed installation and commissioning of GPS-enabled hardware and RF communication systems across domestic and international sites.
- **RF System Integration & Optimization:** Configured and maintained radio repeaters and wireless communication infrastructure to ensure reliable signal coverage in complex environments.

- **System Maintenance & Reliability:** Performed preventive maintenance, system upgrades, and hardware repairs to maintain 24/7 operation of field-deployed assets.
- **Field Diagnostics & Troubleshooting:** Diagnosed and resolved hardware, connectivity, and signal integrity issues in live environments, ensuring minimal system downtime.
- **Customer Training & Support:** Delivered on-site technical training to operators, translating system functionality into practical usage and troubleshooting procedures.

*Field Service Technician | *Southeast U.S., LATAM, Caribbean Territories* | April 2004 – May 2006

- **On-Site Hardware Installation:** Installed GPS tracking systems and supporting network infrastructure on mobile platforms, including wiring, mounting, and system configuration.
 - **Global Field Operations:** Executed deployments across domestic and international locations, managing travel logistics, site preparation, and system installation.
 - **Systems Integration:** Integrated mobile hardware systems with backend platforms, ensuring reliable communication and data transmission.
 - **Field Troubleshooting & Repair:** Diagnosed and repaired hardware failures in rugged outdoor environments, including electrical, connectivity, and component-level issues.
 - **Site Surveys & Deployment Planning:** Assessed installation environments to determine optimal system configuration, power requirements, and signal coverage.
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Education

Bachelor of Science, Computer Engineering

Florida Atlantic University
Boca Raton, FL